C++ Review DUNE 💻

Una organización donde compartir notas acerca de C++ con PDFs escritos en LaTeX.

22 de enero del 2022

📃 Pad de apuntes

📎 Liga del PDF

🎬 Sesión grabada en `diode.zone`

📅 Cronograma de esta semana

$ date (🇵🇪 Lima, 🇨🇴 Bogotá, 🇲🇽 Ciudad de México -1)

- `Sat Jan 22 07:00:00 AM -05 2022`.
- `Sun Jan 23 07:00:00 AM -05 2022`.

git 📓 **https://man.archlinux.org/man/git.1**

git-config 📓 **https://man.archlinux.org/man/git-config.1**

git-init 📓 **https://man.archlinux.org/man/git-init.1.en**

git-remote 📓 **https://man.archlinux.org/man/extra/git/git-remote.1.en**

git-push 📓 **https://man.archlinux.org/man/git-push.1**

git-stash 📓 **https://man.archlinux.org/man/git-stash.1**

git-log 📓 **https://man.archlinux.org/man/git-log.1**

git-show 📓 **https://man.archlinux.org/man/git-show.1**

git-diff 📓 **https://man.archlinux.org/man/git-diff.1**

git-gc 📓 **https://man.archlinux.org/man/extra/git/git-gc.1.en**

git-mv 📓 **https://man.archlinux.org/man/extra/git/git-mv.1.en**

git-rm 📓 **https://man.archlinux.org/man/extra/git/git-rm.1.en**

git-add 📓 **https://man.archlinux.org/man/extra/git/git-add.1.en**

git-grep 📓 **https://man.archlinux.org/man/extra/git/git-grep.1.en**

git-branch 📓 **https://man.archlinux.org/man/git-branch.1**

git-checkout 📓 **https://man.archlinux.org/man/git-checkout.1**

git-rebase 📓 **https://man.archlinux.org/man/extra/git/git-rebase.1.en**

git-lfs 📓 **https://man.archlinux.org/man/community/git-lfs/git-lfs.1.en**

git-log 📓 **https://man.archlinux.org/man/extra/git/git-log.1.en**

git-tag 📓 **https://man.archlinux.org/man/extra/git/git-tag.1.en**

git-diff 📓 **https://man.archlinux.org/man/extra/git/git-diff.1.en**

Actividades

1. Descargue el tarball del repositorio `dune-pdelab`.

```
$ curl -O https://gitlab.dune-project.org/pdelab/dune-pdelab/-/archive/master/dune-pdelab-master.tar.gz
```

2. Realize el tutorial de `git` `https://git-scm.com/docs/gittutorial`.
3. Instale asciinema y regístrese `https://asciinema.org`.

```
$ sudo pacman -S asciinema
# suba las soluciones a, por ejemplo
# https://asciinema.org/a/L7v4biUA6A7m3UNsE28UZy8WK
```

4. Rehacer los comandos de la sesión pasada.

Ejemplos

```
git config
git config --list
git config --global user.name "Name"
git config --list
cd
whoami
mkdir -p Git
ls
pwd
# Primero ubicarse en el home del usuario /home/usuario
echo $USER
whoami
cd
cd /
pwd
ls
cd
pwd
mkdir -p Git/Hub
ls
ls Git/
cd Git/Hub/
ls
git init
git status
git config --global init.defaultBranch main
ls
git status
cd ..
ls
mkdir usuario
```

Ejemplos

```
cd usuario
git init
git status
cd ..
ls
mkdir ramas
cd ramas/
pwd
cd
ls
cd Git/
ls
cd Hub/
git status
cd ..
ls
cd usuario
git status
cd ..
cd ramas/
git status
git init -b rama
git status
#origin/rama
#origin/master
#origin/main
sudo pacman -Sy
pacman -Ss github-cli
pacman -Si github-cli
git clone https://github.com/numpy/numpy.git
ls
```

Ejemplos

```
cd numpy
git branch
git branch --list
git branch -a
pwd
cd
ls
pwd
git status
git branch -a
ls
mkdir src
touch README.md
git status
git add README.md
git add src
git status
git commit
ls -l
# r w x 4 2 1
# 6 4 4
ls -l src/
ls -lh src/
git status
git commit
git add .gitignore
git commit -m "Add .gitignore"
git status
git add src/vectores.cc
git commit -m "Create vectores.cc"
git status
```

Ejemplos

```
git log
HASH_COMMIT=f5a1124eba1bf89e2baa65fc483b4b76146ae9be
git show $HASH_COMMIT^!
git status
git add src/vectores.cc
git commit -m "Change to float from integers"
git log
```

Ejemplos

```cpp
#include <iostream>

int main(int argc, char **argv)
{

  std::cout << "¡Hola mundo!" << std::endl;

  return 0;
}
```

Ejemplos

```cpp
#include <vector>
#include <iostream>

int main(int argc, char **argv)
{
  auto v = std::vector<float>{1.0, 2.0, 3.0, 4.0, 5.0, 6.0, 7.0};

  for (const auto &e : v)
  {
    std::cout << e << std::endl;
  }

  return 0;
}
```

Ejemplos

```cpp
#include <string>
#include <iostream>

int main(int argc, char **argv)
{

  std::string cadena = "hola cadena";
  std::cout << cadena << std::endl;

  return 0;
}
```